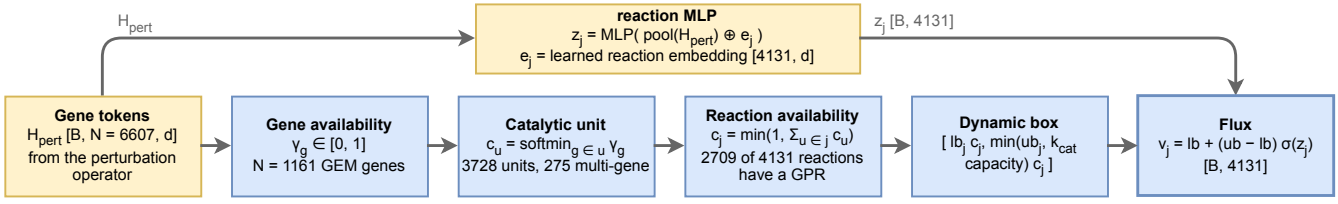
A differentiable enzyme-constrained thermodynamic flux layer

a **Availability to box to flux. A deletion acts on the bounds, so the box is exact and enzyme capacity never needs a penalty.**



one token per gene; only the 1161 GEM genes are read by the layer

deletion sets **$\gamma = 0$ HARD**; every other gene: **learned sigmoid**

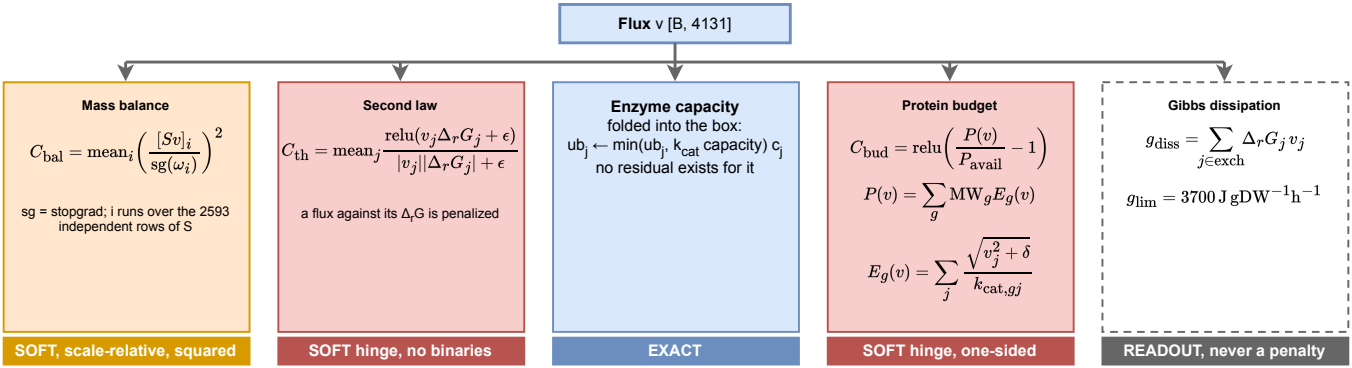
complex: **min** (the scarcest subunit limits)

isozymes: **sum**; reactions without a GPR keep $c_j = 1$

enzyme capacity enters **HERE, exactly**, not as a penalty

the box holds **by construction**, so there is no C_{box} loss term

b **The constraint residuals hanging off v. One is exact by construction; the other four are soft, and none needs a binary variable.**



213 of the 2806 rows of S are redundant; the mean skips them

Thermo-Flux uses one binary variable per reaction; the hinge uses none

a bound (panel a, dynamic box), so v cannot violate it

no reward for approaching P_{avail}; only exceeding it costs

g_{lim} from Niebel 2019, an empirical plateau, not a law

Thermodynamic decomposition of $\Delta_r G_j$, the driving force used by the second law and by dissipation

$$\Delta_r G_j = \Delta_r G'^{\circ}_j + RT \sum_i S_{ij} \ln c_i + \Delta_r G^t_j$$

tabulated: $\Delta_f G'^{0}$ known for 85.1% of metabolites

learned ln c_i, bounded to 0.1 µM to 10 mM; anchored where a measurement exists

transport term: recoverable for 441 transport reactions from the GEM's table; **off by default**

What rests on data, and how much of it is there

S	100%
$\Delta_f G'^{0}$	85.1% of metabolites
MW	100%
k_{cat}	4.0%: 148 of 3728 catalytic units resolved from the Open Enzyme Database
conc. anchors	6.8% of metabolites

exact / structural | soft, squared, scale-relative | soft, one-sided hinge | data-backed quantity | learned | default, or a gap in the data